Code No: 136FD **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, July - 2023
GEOINFORMATICS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a) What lens is best for aerial photography and why? [2]
b) How many minimum numbers of photographs are required for Stereo Photogrammetry and why? [3]
c) What remote sensing does? [2]
d) Write the special needs of sensors for geological studies. [3]
e) What are the features of image histogram? [2]
f) Distinguish between supervised and unsupervised classification. [3]
g) How GIS works in urban planning? [2]
h) List kind of errors that can occur in GIS. [3]
i) What are GNSS devices? [2]
j) What do you understand by GALILEO? [3]

PART - B

(50 Marks)

2. What is aerial photogrammetry? Discuss the various types of aerial photographs based on (a) camera axis, (b) the scale of the image and (c) the type of film used. [10]

OR

3.a) Explain briefly about digital photogrammetry.
b) Write a short note on Relief Displacement. [5+5]

4. What is the principle and process of remote sensing? Discuss what type of sensors are used in satellites? [10]

OR

5.a) Explain about physics of Remote Sensing.
b) Discuss about Remote Sensing satellites and their products. [5+5]

6.a) Explain crop inventory using remote sensing.
b) Explain the remote sensing application in land use and land cover studies. [5+5]

OR

7. What are the different methods of LULC classification systems? Discuss their importance with suitable examples. [10]

8.a) Classify data in GIS context and explain spatial data editing.
b) Describe the theoretical frame work and operations involved in Geographic Information System. [5+5]

OR

9.a) Discuss the advantages and disadvantages of raster data model when compared with vector data.
b) What are the steps involved in GIS process? Explain types of Spatial Analysis. [5+5]

10. How the Global Positioning System works? Explain the Space and Control segments in GPS. [10]

OR

11.a) Give the details of the important map projections applicable to Indian regions.
b) Explain in detail the applications of GPS. [5+5]

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